MIMIC

caregiver

A description table for the ICU caregivers in the ICU module referenced by *caregiver_id*. As of MIMIC-IV v2.2, this table simply lists all unique `caregiver_id` in the database.

Note that, in order to distinguish identifiers used in the hospital wide EHR from those used in the ICU information system, we have adopted the nomenclature of "caregivers" for the ICU (`caregiver_id` and *caregivers*). For the hospital data in the hosp module, we use the terminology of "providers" (`provider_id` and *providers*). However, conceptually, both these sets of identifiers and tables refer to practicing providers at the hospital.

Table columns

Name	Postgres data type
caregiver_id	VARCHAR(10) NOT NULL

`caregiver_id`

`caregiver_id` lists all possible identifiers for caregivers used in the ICU module. `caregiver_id` uniquely identifies a single caregiver who documented data in the ICU information system.

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MIMIC

The chartevents table

Table source: MetaVision ICU database.

Table purpose: Contains all charted data for all patients.

Number of rows: 313,645,063

Links to:

- patients on `subject_id`
- admissions on `hadm_id`
- icustays on `stay_id`
- d_items on `itemid`

Brief summary

chartevents contains all the charted data available for a patient. During their ICU stay, the primary repository of a patient's information is their electronic chart. The electronic chart displays patients' routine vital signs and any additional information relevant to their care: ventilator settings, laboratory values, code status, mental status, and so on. As a result, the bulk of information about a patient's stay is contained in *chartevents*. Furthermore, even though laboratory values are captured elsewhere (*labevents*), they are frequently repeated within *chartevents*. This occurs because it is desirable to display the laboratory values on the patient's electronic chart, and so the values are copied from the database storing laboratory values to the database storing the *chartevents*.

Important considerations

- Some items are duplicated between the *labevents* and *chartevents* tables. In cases where there is disagreement between measurements, *labevents* should be taken as the ground truth.

Table columns

Name	Postgres Data type
subject_id	INTEGER
hadm_id	INTEGER
stay_id	INTEGER
caregiver_id	INTEGER
charttime	TIMESTAMP(0)
storetime	TIMESTAMP(0)
itemid	INTEGER
value	VARCHAR(200)
valuenum	DOUBLE PRECISION
valueuom	VARCHAR(20)
warning	SMALLINT

`subject_id`, `hadm_id`, `stay_id`

Identifiers which specify the patient: `subject_id` is unique to a patient, `hadm_id` is unique to a patient hospital stay and `stay_id` is unique to a patient ward stay. More information about these identifiers is available here.

`caregiver_id`

`caregiver_id` uniquely identifies a single caregiver who documented data in the ICU

information system.

`charttime`, `storetime`

`charttime` records the time at which an observation was made, and is usually the closest proxy to the time the data was actually measured. `storetime` records the time at which an observation was manually input or manually validated by a member of the clinical staff.

`itemid`

Identifier for a single measurement type in the database. Each row associated with one `itemid` (e.g. 220045) corresponds to an instantiation of the same measurement (e.g. heart rate).

`value`, `valuenum`

`value` contains the value measured for the concept identified by the `itemid`. If this value is numeric, then `valuenum` contains the same data in a numeric format. If this data is not numeric, `valuenum` is null. In some cases (e.g. scores like Glasgow Coma Scale, Richmond Sedation Agitation Scale and Code Status), `valuenum` contains the score and `value` contains the score and text describing the meaning of the score.

`valueuom`

`valueuom` is the unit of measurement for the `value`, if appropriate.

`warning`

`warning` specifies if a warning for this observation was manually documented by the care provider.

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MIMIC

The d_items table

Table source: Metavision ICU databases.

Table purpose: Definition table for all items in the ICU databases.

Number of rows: 4,014

Links to:

- *chartevents* on `itemid`
- *datetimeevents* on `itemid`
- *inputevents* on `itemid`
- *outputevents* on `itemid`
- *procedureevents* on `itemid`

Important considerations

- If the `linksto` column is null, then the data is currently unavailable, but planned for a future release.

Table columns

Name	Postgres data type
itemid	INTEGER
label	VARCHAR(200)
abbreviation	VARCHAR(100)

linksto	VARCHAR(50)
category	VARCHAR(100)
unitname	VARCHAR(100)
param_type	VARCHAR(30)
lownormalvalue	FLOAT
highnormalvalue	FLOAT

Detailed Description

The D_ITEMS table defines `itemid`, which represents measurements in the database. Measurements of the same type (e.g. heart rate) will have the same `itemid` (e.g. 220045). Values in the `itemid` column are unique to each row. All `itemid`s will have a value > 220000.

`itemid`

As an alternate primary key to the table, `itemid` is unique to each row.

`label`, `abbreviation`

The `label` column describes the concept which is represented by the `itemid`. The `abbreviation` column, only available in Metavision, lists a common abbreviation for the label.

`linksto`

`linksto` provides the table name which the data links to. For example, a value of 'chartevents' indicates that the `itemid` of the given row is contained in CHARTEVENTS. A single `itemid` is only used in one event table, that is, if an `itemid` is contained in CHARTEVENTS it will *not* be contained in any other event table (e.g. IOEVENTS, CHARTEVENTS, etc).

`category`

`category` provides some information of the type of data the `itemid` corresponds to. Examples include 'ABG', which indicates the measurement is sourced from an arterial blood gas, 'IV Medication', which indicates that the medication is administered through an intravenous line, and so on.

`unitname`

`unitname` specifies the unit of measurement used for the `itemid`. This column is not always available, and this may be because the unit of measurement varies, a unit of measurement does not make sense for the given data type, or the unit of measurement is simply missing. Note that there is sometimes additional information on the unit of measurement in the associated event table, e.g. the `valueuom` column in CHARTEVENTS.

`param_type`

`param_type` describes the type of data which is recorded: a date, a number or a text field.

`lownormalvalue`, `highnormalvalue`

These columns store reference ranges for the measurement. Note that a reference range encompasses the *expected* value of a measurement: values outside of this may still be physiologically plausible, but are considered unusual.

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MIMIC

MIMIC-IV / ICU / datetimeevents

The datetimeevents table

Table source: MetaVision ICU database.

Table purpose: Contains all date formatted data.

Number of rows: 7,112,999

Links to:

- patients on `subject_id`
- admissions on `hadm_id`
- icustays on `stay_id`
- d_items on `itemid`

Table columns

Name	Postgres Data type
subject_id	INTEGER
hadm_id	INTEGER
stay_id	INTEGER
caregiver_id	INTEGER
charttime	TIMESTAMP(3)
storetime	TIMESTAMP(3)
itemid	INTEGER
value	TIMESTAMP(3)

valueuom	VARCHAR(20)
warning	SMALLINT

Detailed Description

datetimeevents contains all date measurements about a patient in the ICU. For example, the date of last dialysis would be in the *datetimeevents* table, but the systolic blood pressure would not be in this table. As all dates in MIMIC are anonymized to protect patient confidentiality, all dates in this table have been shifted. Note that the chronology for an individual patient has been unaffected however, and quantities such as the difference between two dates remain true to reality.

`subject_id`, `hadm_id`, `stay_id`

Identifiers which specify the patient: `subject_id` is unique to a patient, `hadm_id` is unique to a patient hospital stay and `stay_id` is unique to a patient ward stay.

`caregiver_id`

`caregiver_id` uniquely identifies a single caregiver who documented data in the ICU information system.

`charttime`, `storetime`

`charttime` records the time at which an observation was charted, and is usually the closest proxy to the time the data was actually measured. `storetime` records the time at which an observation was manually input or manually validated by a member of the clinical staff.

`itemid`

Identifier for a single measurement type in the database. Each row associated with one `itemid` (e.g. 212) corresponds to an instantiation of the same measurement (e.g. heart rate).

`value`

The documented date - this is the value that corresponds to the concept referred to by `itemid`. For example, if querying for `itemid`: 225755 (“18 Gauge Insertion Date”), then the `value` column indicates the date the line was inserted.

`valueuom`

The unit of measurement for the value - almost always the text string “Date”.

`warning`

`warning` specifies if a warning for this observation was manually documented by the care provider.

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The icustays table

Table source: Hospital database.

Table purpose: Defines each ICU stay in the database using STAY_ID.

Number of rows: 73,181

Links to:

- patients on `subject_id`
- admissions on `hadm_id`

Important considerations

- `stay_id` is a *generated* identifier that is *not* based on any raw data identifier. The hospital and ICU databases are not intrinsically linked and so do not have any concept of an ICU encounter identifier.
- The `icustays` table is derived from the `transfers` table. Specifically, it excludes rows in `transfers` where the ward is not an ICU.
- Multiple consecutive icu stay entries from `transfers` are merged into a single entry in `transfers`.
- Heart rate measurements are used to determine if an icu stay is valid. If the heart rate measurement for an icu stay in `transfers` is not available, the icu stay will not be included in `icustays` (~5%).

Table columns

Name	Postgres data type

subject_id	INT
hadm_id	INT
stay_id	INT
first_careunit	VARCHAR(20)
last_careunit	VARCHAR(20)
intime	TIMESTAMP(0)
outtime	TIMESTAMP(0)
los	DOUBLE PRECISION

Detailed Description

`subject_id`, `hadm_id`, `stay_id`

Identifiers which specify the patient: `subject_id` is unique to a patient, `hadm_id` is unique to a patient hospital stay and `stay_id` is unique to a patient ward stay.

`FIRST_CAREUNIT`, `LAST_CAREUNIT`

`FIRST_CAREUNIT` and `LAST_CAREUNIT` contain, respectively, the first and last ICU type in which the patient was cared for. As an `stay_id` groups all ICU admissions within 24 hours of each other, it is possible for a patient to be transferred from one type of ICU to another and have the same `stay_id`.

Care units are derived from the TRANSFERS table, and definition for the abbreviations can be found in the documentation for TRANSFERS.

`INTIME`, `OUTTIME`

`INTIME` provides the date and time the patient was transferred into the ICU. `OUTTIME` provides the date and time the patient was transferred out of the ICU.

LOS

`LOS` is the length of stay for the patient for the given ICU stay, which may include one or more ICU units. The length of stay is measured in fractional days.

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MIMIC

MIMIC-IV / ICU / Ingredientevents

The *inputevents* table

Table source: MetaVision ICU database.

Table purpose: Ingredients contained within inputs data for patients.

Number of rows: 12,229,408

Links to:

- patients on `subject_id`
- admissions on `hadm_id`
- icustays on `stay_id`
- d_items on `itemid`

Important considerations

- In the source data, there is no concept of a volume in the database: only a `rate`. All rows are recorded with a `starttime` and an `endtime`. The `amount` in this table is *derived* from the `rate` column, and provided for convenience. An exception is bolus administrations: these are listed as ending one minute after they started, i.e. `endtime`: `starttime` + 1 minute, and they do not have a `rate`.

Table columns

Name	Postgres data type
subject_id	INTEGER
hadm_id	INTEGER

stay_id	INTEGER
caregiver_id	INTEGER
starttime	TIMESTAMP(0)
endtime	TIMESTAMP(0)
storetime	TIMESTAMP(0)
itemid	INTEGER
amount	DOUBLE PRECISION
amountuom	VARCHAR(20)
rate	DOUBLE PRECISION
rateuom	VARCHAR(20)
orderid	INTEGER
linkorderid	INTEGER
statusdescription	VARCHAR(20)
originalamount	DOUBLE PRECISION
originalrate	DOUBLE PRECISION

Detailed Description

`subject_id`, `hadm_id`, `stay_id`

Identifiers which specify the patient: `subject_id` is unique to a patient, `hadm_id` is unique to a patient hospital stay and `stay_id` is unique to a patient ICU stay.

`caregiver_id`

`caregiver_id` uniquely identifies a single caregiver who documented data in the ICU

information system.

`starttime`, `endtime`

`starttime` and `endtime` record the start and end time of the event.

`storetime`

`storetime` records the time at which an observation was manually input or manually validated by a member of the clinical staff.

`itemid`

Identifier for a single measurement type in the database. Each row associated with one `itemid` which corresponds to an instantiation of the same measurement (e.g. norepinephrine).

`amount`, `amountuom`

`amount` and `amountuom` list the amount of a drug or substance administered to the patient either between the `starttime` and `endtime`.

`rate`, `rateuom`

`rate` and `rateuom` list the rate at which the drug or substance was administered to the patient either between the `starttime` and `endtime`.

`orderid`, `linkorderid`

`orderid` links multiple items contained in the same solution together. For example, when a solution of noradrenaline and normal saline is administered both noradrenaline and normal saline occur on distinct rows but will have the same `orderid`.

`linkorderid` links the same order across multiple instantiations: for example, if the rate of delivery for the solution with noradrenaline and normal saline is changed, two new rows which share the same new `orderid` will be generated, but the `linkorderid` will be the same.

`statusdescription`

`statusdescription` states the ultimate status of the item, or more specifically, row. It is used to indicate why the delivery of the compound has ended. There are only six possible statuses:

- Changed - The current delivery has ended as some aspect of it has changed (most frequently, the rate has been changed)
- Paused - The current delivery has been paused
- FinishedRunning - The delivery of the item has finished (most frequently, the bag containing the compound is empty)
- Stopped - The delivery of the item been terminated by the caregiver
- Rewritten - Incorrect information was input, and so the information in this row was rewritten (these rows are primarily useful for auditing purposes - the rates/amounts described were *not* delivered and so should not be used if determining what compounds a patient has received)
- Flushed - A line was flushed.

`originalamount`

Drugs are usually mixed within a solution and delivered continuously from the same bag. This column represents the amount of the compound contained in the bag at `starttime`.

`originalrate`

This is the rate that was input by the care provider. Note that this may differ from `rate` because of various reasons: `originalrate` was the original planned rate, while the `rate` column will be the true rate delivered.

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MIMIC

The *inputevents* table

Table source: MetaVision ICU database.

Table purpose: Input data for patients.

Number of rows: 8,978,893

Links to:

- patients on `subject_id`
- admissions on `hadm_id`
- icustays on `stay_id`
- d_items on `itemid`

Brief example

The original source database recorded input data using two tables: RANGESIGNALS and ORDERENTRY. These tables do not appear in MIMIC as they have been merged to form the INPUTEVENTS table. RANGESIGNALS contains recorded data elements which last for a fixed period of time. Furthermore, the RANGESIGNALS table recorded information for each component of the drug separately. For example, for a norepinephrine administration there would be two components: a main order component (norepinephrine) and a solution component (NaCl). The `starttime` and `endtime` of RANGESIGNALS indicated when the drug started and finished. *Any* change in the drug rate would result in the current infusion ending, and a new `starttime` being created.

Let's examine an example of a patient being given norepinephrine.

Item	`starttime`	`endtime`	`rate`	`rateuom`	

Norepinephrine	18:20	18:25	1	mcg/kg/min	8
NaCl	18:20	18:25	10	ml/hr	8
Norepinephrine	18:25	20:00	2	mcg/kg/min	8
NaCl	18:25	20:00	20	ml/hr	8

The `starttime` for the solution (NaCl) and the drug (norepinephrine) would be 18:20. The rate of the drug is 1 mcg/kg/min, and the rate of the solution is 10 mL/hr. The nurse decides to increase the drug rate at 18:25 to 2 mcg/kg/min. As a result, the `endtime` for the two rows corresponding to the solution (NaCl and norepinephrine) is set to 18:25. Two new rows are generated with a `starttime` of 18:25. These two new rows would continue until either (i) the drug rate was changed or (ii) the drug was delivery was discontinued. The `orderid` column is used to group drug delivery with rate of delivery. In this case, we have NaCl and norepinephrine in the same bag delivered at the same time - as a result their `orderid` is the same (8003). When the rate is changed, a new `orderid` is generated (8020). The column `linkorderid` can be used to link this drug across all administrations, even when the rate is changed. Note also that `linkorderid` is always equal to the first `orderid` which occurs for the solution, as demonstrated in the example above.

Important considerations

- For Metavision data, there is no concept of a volume in the database: only a `rate`. All inputs are recorded with a `starttime` and an `endtime`. As a result, the volumes in the database for Metavision patients are *derived* from the rates. Furthermore, exact start and stop times for the drugs are easily deducible.
- A bolus will be listed as ending one minute after it started, i.e. `endtime`: `starttime` + 1 minute

Table columns

Name	Postgres data type

subject_id	INT
hadm_id	INT
stay_id	INT
caregiver_id	INTEGER
starttime	TIMESTAMP(0)
endtime	TIMESTAMP(0)
storetime	TIMESTAMP(0)
itemid	INT
amount	DOUBLE PRECISION
amountuom	VARCHAR(30)
rate	DOUBLE PRECISION
rateuom	VARCHAR(30)
orderid	BIGINT
linkorderid	BIGINT
ordercategoryname	VARCHAR(100)
secondaryordercategoryname	VARCHAR(100)
ordercomponenttypedescription	VARCHAR(200)
ordercategorydescription	VARCHAR(50)
patientweight	DOUBLE PRECISION
totalamount	DOUBLE PRECISION
totalamountuom	VARCHAR(50)
`isopenbag`	SMALLINT

statusdescription	VARCHAR(30)
originalamount	DOUBLE PRECISION
originalrate	DOUBLE PRECISION

Detailed Description

`subject_id`, `hadm_id`, `stay_id`

Identifiers which specify the patient: `subject_id` is unique to a patient, `hadm_id` is unique to a patient hospital stay and `stay_id` is unique to a patient ICU stay.

`caregiver_id`

`caregiver_id` uniquely identifies a single caregiver who documented data in the ICU information system.

`starttime`, `endtime`

`starttime` and `endtime` record the start and end time of an input/output event.

`storetime`

`storetime` records the time at which an observation was manually input or manually validated by a member of the clinical staff.

`itemid`

Identifier for a single measurement type in the database. Each row associated with one `itemid` which corresponds to an instantiation of the same measurement (e.g. norepinephrine).

`amount`, `amountuom`

`amount` and `amountuom` list the amount of a drug or substance administered to the patient

either between the `starttime` and `endtime`.

rate, rateuom

`rate` and `rateuom` list the rate at which the drug or substance was administered to the patient either between the `starttime` and `endtime`.

orderid, linkorderid

`orderid` links multiple items contained in the same solution together. For example, when a solution of noradrenaline and normal saline is administered both noradrenaline and normal saline occur on distinct rows but will have the same `orderid`.

`linkorderid` links the same order across multiple instantiations: for example, if the rate of delivery for the solution with noradrenaline and normal saline is changed, two new rows which share the same new `orderid` will be generated, but the `linkorderid` will be the same.

`ordercategoryname`, `secondaryordercategoryname`, `ordercomponenttypedescription`, `ordercategorydescription`

These columns provide higher level information about the order the medication/solution is a part of. Categories represent the type of administration, while the `ordercomponenttypedescription` describes the role of the substance in the solution (i.e. main order parameter, additive, or mixed solution)

`patientweight`

The patient weight in kilograms.

`totalamount`, `totalamountuom`

Intravenous administrations are usually given by hanging a bag of fluid at the bedside for continuous infusion over a certain period of time. These columns list the total amount of the fluid in the bag containing the solution.

`isopenbag`

Whether the order was from an open bag.

`continueinnextdept`

If the order ended on patient transfer, this field indicates if it continued into the next department (e.g. a floor).

`statusdescription`

`statusdescription` states the ultimate status of the item, or more specifically, row. It is used to indicate why the delivery of the compound has ended. There are only six possible statuses:

- Changed - The current delivery has ended as some aspect of it has changed (most frequently, the rate has been changed)
- Paused - The current delivery has been paused
- FinishedRunning - The delivery of the item has finished (most frequently, the bag containing the compound is empty)
- Stopped - The delivery of the item been terminated by the caregiver
- Flushed - A line was flushed.

`originalamount`

Drugs are usually mixed within a solution and delivered continuously from the same bag. This column represents the amount of the drug contained in the bag at `starttime`. For the first infusion of a new bag, `originalamount`: `totalamount`. Later on, if the rate is changed, then the amount of the drug in the bag will be lower (as some has been administered to the patient). As a result, `originalamount` < `totalamount`, and `originalamount` will be the amount of drug leftover in the bag at that `starttime`.

`originalrate`

This is the rate that was input by the care provider. Note that this may differ from `rate` because of various reasons: `originalrate` was the original planned rate, while the `rate` column will be the true rate delivered. For example, if a a bag is about to run out and

the care giver decides to push the rest of the fluid, then `rate` > `originalrate`. However, these two columns are usually the same, but have minor non-clinically significant differences due to rounding error.

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MIMIC

MIMIC-IV / ICU / outputevents

The *outputevents* table

Table source: MetaVision ICU database.

Table purpose: Output data for patients.

Number of rows: 4,234,967

Links to:

- patients on `subject_id`
- admissions on `hadm_id`
- icustays on `stay_id`
- d_items on `itemid`

Table columns

Name	Data type
subject_id	INTEGER
hadm_id	INTEGER
stay_id	INTEGER
caregiver_id	INTEGER
charttime	TIMESTAMP(3)
storetime	TIMESTAMP(3)
itemid	INTEGER
value	DOUBLE PRECISION

valueuom	VARCHAR(20)

Detailed Description

`subject_id`, `hadm_id`, `stay_id`

Identifiers which specify the patient: `subject_id` is unique to a patient, `hadm_id` is unique to a patient hospital stay and `stay_id` is unique to a patient ICU stay.

`caregiver_id`

`caregiver_id` uniquely identifies a single caregiver who documented data in the ICU information system.

`charttime`

`charttime` is the time of an output event.

`storetime`

`storetime` records the time at which an observation was manually input or manually validated by a member of the clinical staff.

`itemid`

Identifier for a single measurement type in the database. Each row associated with one `itemid` (e.g. 212) corresponds to an instantiation of the same measurement (e.g. heart rate).

`value`, `valueuom`

`value` and `valueuom` list the amount of a substance at the `charttime` (when the exact start time is unknown, but usually up to an hour before).

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MIMIC

MIMIC-IV / ICU / procedureevents

The procedureevents table

Table source: MetaVision ICU database.

Table purpose: Contains procedures for patients

Number of rows: 696,092

Links to:

- patients on `subject_id`
- admissions on `hadm_id`
- icustays on `stay_id`
- d_items on `itemid`

Important considerations

This table is **not** a required documentation field during routine care. As a result, existence of procedures here indicates their presence, but absence does not indicate the procedure was not conducted. The consistency of documentation varies by the type of procedure. For example, invasive ventilation tends to be documented, whereas non-invasive documentation is less consistently documented.

Table columns

Name	Data type
subject_id	INTEGER
hadm_id	INTEGER

stay_id	INTEGER
caregiver_id	INTEGER
starttime	TIMESTAMP
endtime	TIMESTAMP
storetime	TIMESTAMP
itemid	INTEGER
value	DOUBLE PRECISION
valueuom	VARCHAR(20)
location	VARCHAR(100)
locationcategory	VARCHAR(50)
orderid	INTEGER
linkorderid	INTEGER
ordercategoryname	VARCHAR(50)
ordercategorydescription	VARCHAR(30)
patientweight	DOUBLE PRECISION
isopenbag	SMALLINT
continueinnextdept	SMALLINT
statusdescription	VARCHAR(20)
originalamount	DOUBLE PRECISION
originalrate	DOUBLE PRECISION

Detailed Description

`subject_id`, `hadm_id`, `stay_id`

Identifiers which specify the patient: `subject_id` is unique to a patient, `hadm_id` is unique to a patient hospital stay and `stay_id` is unique to a patient ICU stay.

`caregiver_id`

`caregiver_id` uniquely identifies a single caregiver who documented data in the ICU information system.

`starttime`, `endtime`

`starttime` and `endtime` record the start and end time of an event.

`storetime`

`storetime` specifies the time when the event was recorded in the system.

`itemid`

Identifier for a single measurement type in the database. Each row associated with one `itemid` (e.g. 212) corresponds to a type of measurement (e.g. heart rate). The *d_items* table may be joined on this field. For any itemid appearing in the *procedureevents* table, *d_items* `linksto` column will have the value 'procedureevents'.

`value`

In the `procedureevents` table, this identifies the duration of the procedure (if applicable). For example, if querying for itemid 225794 ("Non-invasive Ventilation"), then the value column indicates the duration of ventilation therapy.

`valueuom`

The unit of measurement for the value. Most frequently "None" (no value recorded); otherwise one of "day", "hour", or "min". A query for itemiid 225794 ("Non-invasive

Ventilation") returning a `value` of 461 and `valueuom` of 'min' would correspond to non-invasive ventilation provided for 461 minutes; this value is expected to match the difference between the `starttime` and `endtime` fields for the record. A procedure with `valueuom` equal to "None" corresponds to a procedure which is instantaneous (e.g. intubation, patient transfer) or whose duration is not relevant (e.g. imaging procedures). For these records, there will be a difference of one second between `starttime` and `endtime` values.

`location`, `locationcategory`

`location` and `locationcategory` provide information about where on the patient's body the procedure is taking place. For example, the `location` might be 'Left Upper Arm' and the `locationcategory` might be 'Invasive Venous'.

`orderid`, `linkorderid`

These columns link procedures to specific physician orders. Unlike in the `mimic_icu.inputevents` table, most procedures in `procedureevents` are ordered independently.

There are a limited number of records for which the same procedure was performed again at a later date under the same original order. When a procedure was repeated under the same original order, the `linkorderid` field of the record for the later procedure will be set to the `orderid` field of the earlier record. In all other cases, `orderid` = `linkorderid`.

`ordercategoryname`, `ordercategorydescription`

These columns provide higher level information about the medication/solution order. Categories represent the type of administration.

`patientweight`

The patient weight in kilograms.

`isopenbag`

Whether the order was from an open bag.

`continueinnextdept`

If the order ended on patient transfer, this field indicates if it continued into the next department (e.g. a floor).

`statusdescription`

`statusdescription` states the ultimate status of the procedure referred to in the row. The statuses appearing on the `procedureevents` table are:

- `Paused` - The current delivery has been paused.
- `FinishedRunning` - The delivery of the item has finished (most frequently, the bag containing the compound is empty).
- `Stopped` - The delivery of the item been terminated by the caregiver.

Nearly all procedures recorded in *procedureevents* have a status of `FinishedRunning`.

`originalamount`, `originalrate`

These fields are present in the table and never null, but have no clear meaning. In particular, “originalrate” is either 0 or 1 for all records.

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